Shreehan Santosh Kate

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Portfolio

EDUCATION

2022/11 – Present
Trichy, India

B.Tech - Instrumentation and Control Engineering
National Institute of Technology, Tiruchirappalli
CGPA: 8/10

CONFERENCE PUBLICATIONS

Adaptive PID Control for Quadrotor UAVs Using Feedback Linearization and Deep Reinforcement Learning (Submitted)
Shreehan Santosh Kate, *T. Shuprajhaa, K. Srinivasan*
2026 IEEE North-East India International Energy Conversion Conference and Exhibition

Deep Reinforcement Learning Framework for Adaptive Control of Swing-up Inverted Pendulum (Accepted)
Shreehan Santosh Kate, *T. Shuprajhaa, K. Srinivasan*
5th International Conference on Emerging Electronics and Automation

Bio-Inspired Optimization of LQR Controllers for Feedback-Linearized Cruise Missiles
Chitta Subbareddy, Ramakalyan Ayyagari, ***Shreehan Santosh Kate***
9th International Conference on Inventive Systems and Control

RESEARCH EXPERIENCE

2024/02 – 2025/09
Trichy, India

National Institute of Technology, Tiruchirappalli
Guide: *Prof. Dr. Ramakalyan Ayyagari and Prof. Subbareddy Chitta*

- Developed a **bio-inspired** tuning framework for **LQR** Controllers, using **Genetic Algorithm** and **Particle Swarm Optimization**. Achieved **10.4%** faster settling time and **20%** lower RMSE, respectively, than analytical tuning approaches.
- Developed an **Artificial Neural Network (ANN)-based autopilot** control architecture that was trained on flight data to adapt to nonlinear aircraft dynamics. Resulted in a **49.8%** reduction in settling time and a **5%** reduction in peak response.
- Implemented a **data-driven** control strategy for a mobile system by **identifying** a linear state-space model with the **N4SID** algorithm and designed a **PID** controller for stable closed-loop performance

2024/05 – 2024/08
Bengaluru, India

Indian Institute of Science, Bengaluru
Guide: *Prof. Dr. Jishnu Keshavan*

- Designed and motion-studied a **UAV docking mechanism** in **Autodesk Fusion 360** to enable payload lifting in multi-drone operations. The final design was subsequently used for fabrication and assembly.
- Devised a **computer vision**-based approach for **ArUco** marker detection using **OpenCV** and **Python** to facilitate the precise landing of UAVs on moving UGV platforms.
- Estimated the **6-DoF pose** of a camera by implementing the **Perspective-n-Point** (PnP) algorithm in **Python** to compute rotational and translational vectors from 3D world and 2D image points.

2025/05 – 2025/08
Trichy, India

National Institute of Technology, Tiruchirappalli
Guide: *Prof. Dr. K Srinivasan and Prof. Dr. T Shuprajhaa*

- Synthesized a **nonlinear** control law by deriving the inverse dynamical equations of a Quadrotor UAV to cancel nonlinearities for attitude and altitude control.
- Proposed a hybrid deep actor-critic reinforcement learning framework supervising an **Internal Model Controller** for **glycemic regulation**.
- Applied **transfer learning** with a pre-trained **PPO** agent to develop an **adaptive** controller that regulates glycemic levels across diverse **Type-1** Diabetes patient models.

PROFESSIONAL EXPERIENCE

2025/08 – Present
Kanpur, India

xTerra Robotics - IIT Kanpur
Robotics Software Development Intern

- Architected a **joint and Cartesian impedance control** framework in **C++** and **ROS 2** for stable **posture regulation** and compliant **task-space manipulations** for a **5-DOF** robotic manipulator.
- Engineered a residual-based **wrench estimator** that maps manipulator-induced forces to a floating-base whole-body controller to ensure **quadruped stability** during dynamic **loco-manipulation**.
- Led design and development of a kinematic model and performed **link-length optimisation** for a **7-DOF** redundant manipulator intended for space applications.

2025/03 – 2025/07
Bengaluru, India

Bhatiyani Astute Intelligence Limited
Project Intern

- Prepared image and video datasets using **Computer Vision Annotation Tool** for computer vision tasks through **annotation**, **preprocessing**, and **augmentation.**
- Trained and fine-tuned **YOLOv5** models for **object detection** and **classification** of inventory items in warehouses. Evaluated performance using **mAP** and **F1-score** to improve accuracy.

PROJECTS

2025/05 – 2025/08

Reinforcement Learning Framework for Adaptive Control of Quadrotor UAVs

- Developed and trained actor-critic **Twin-Delayed Deep Deterministic Policy Gradient (TD3)** and **Deep Deterministic Policy Gradient (DDPG)** algorithms for dynamic adjustment of controller gains.
- Devised a reward function based on **Linear Quadratic Gaussian** (LQG) cost to guide the agent towards optimal performance through iterative learning.
- Exhibited **disturbance rejection** capabilities; maintained stable tracking under **external perturbations.**
- Achieved faster tracking with **reduced** settling time and **60%** lower cumulative tracking error (IAE, ISE) compared to fixed-gain and model-based controllers (PID, LQR, and LMPC).

2024/03 – 2024/06

NonLinear Control of cruise missiles using fuzzy-tuned LQR

- Derived a **quasi-linearized fourth-order** inner loop model from a nonlinear **longitudinal** missile autopilot model using **Feedback Linearization** to enable linear state-feedback control design.
- Designed a **Mamdani Type-1** fuzzy inference system in **MATLAB** to facilitate **online** adaptation of LQR weights **(Q and R)** in response to varying operational trajectories.
- Enhanced the robustness of the autopilot controller by **28.92%** compared to offline tuning approaches.

2023/08 – 2024/01

All-Terrain Vehicle Design

- Engineered and validated the complete **suspension and steering** systems as part of the development of the all-terrain vehicle. Achieved track width reductions of **3.77% (front)** and **2.13% (rear)**.
- Engineered custom **mounts** and g**u**ssets in **AutoCAD** to ensure the structural integrity and secure integration of the **engine, shocks,** and **suspension arms**.

TECHNICAL SKILLS

Programming Languages
MATLAB, Python, C++

Engineering Software
Simulink, Autodesk Fusion 360, AutoCAD, ROS2, Gazebo

Frameworks
PyTorch, OpenCV, CVAT, NumPy, MuJoCo

Computing and Development Environments
Windows, Google Colab, Jupyter Notebooks, VS Code, Linux

SCHOLASTIC ACHIEVEMENTS AND CERTIFICATIONS

NITT Undergraduate STEM Research Fellowship Awardee
RECAL, The Alumni Association of NIT Trichy
Awarded the **Top Merit** Award and a grant of **Rs. 70,000** at the NITT Undergraduate STEM Research Awards 2024 for a research proposal titled "Adaptive Reinforcement Learning based Predictive Control of Autonomous Quadrotor Systems."

Third Place
Institution's Innovation Council, NIT Trichy
Secured **third** place among 30 teams in **Genesis '23**, the annual startup competition organized by the Institution's Innovation Council, NIT Trichy.

Reinforcement Learning
NPTEL - IIT Madras
Certified for Reinforcement Learning. (Instructor: Dr. B Ravindran)

Drone Systems and Control
NPTEL - IISc Bangalore
Certified in Drone System Design and Control. (Instructor: Dr. Suresh Sundaram)

EXTRACURRICULAR ACTIVITIES AND VOLUNTEERING

2025/07 – 2026/02
NIT Trichy

Chairperson
SENSORS, NIT Trichy

- SENSORS is the **national-leve**l technical symposium of the Instrumentation and Control Department.
- Led planning, coordination, and execution of events and guest lectures, and delivered a high-impact technical and networking platform for students, industry professionals, and researchers.

2023/08 – 2024/08
NIT Trichy

Engineer - Fabrication and Design
PSI Racing Team - NIT Trichy

- Lectured on vehicle dynamics and hands-on fabrication methods for **100+** undergrads.